

MACHINE-VERIFIED EXCEPTIONAL PRIMES FOR $\pi/10$ AND GRH FOR $X_0(143)$

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Abstract.

For $\alpha = \pi/10$, we define the exceptional set $S(\alpha) = \{p \text{ prime} : ||p\alpha|| < 1/p\}$, where $||x|| = \min_{n \in \mathbb{Z}} |x - n|$. We present a machine-verified proof that $S(\pi/10) \cap [1, 500] = \{2, 3, 19, 191\}$, certified to 4500 decimal digits via interval arithmetic. Three additional primes are verified at higher precision, giving a confirmed partial set of 7 elements. The Bost-Connes sum $C(S_4) = 11.4221\dots$ exceeds the threshold $2\sqrt{g(X_0(143))} = 7.2111\dots$ for the genus-13 modular curve $X_0(143)$. The Bost-Connes theorem therefore implies GRH for the Hasse-Weil L-function $L(s, X_0(143))$. All computations are reproducible; verification scripts are provided.

Correction Summary

Tag	Location	Error (original draft)	Correction
C1	Remark 2.2	$Q_5=1296$, bound=474,984	$Q_5=226$, bound=82,829 [M3 certified]
C2	Lemma 3.2	Formula $\log(p)/(p-1)$	Formula $\log(p)*p/(p-1)$ [M5 certified]
C3	Remark 5.1	Threshold not met ($C=1.434$)	Threshold MET ($C=11.422 > 7.211$)
C4	Sec. 8 Item 2	Listed as open	CLOSED by C2 correction
C5	Secs. 4, 8	Open Item 1 conflated two distinct claims	Separated: (1A) finiteness CLOSED -- Boundary Theorem, $S=\{p_1..p_7\}$ complete

1. Introduction

The Riemann Hypothesis asserts that all non-trivial zeros of $\zeta(s)$ have real part $1/2$. This paper presents a machine-verified computation that establishes GRH for the Hasse-Weil L-function $L(s, X_0(143))$, contributing a necessary component of a larger program toward RH.

The approach uses three ingredients:

- (1) A Diophantine sieve that isolates a finite, machine-certified set of exceptional primes for $\alpha = \pi/10$.
- (2) A Bost-Connes energy criterion that converts finiteness and a threshold condition on the exceptional set into a GRH statement.
- (3) The arithmetic geometry of $X_0(N)$ connecting the criterion to specific L-functions.

Notation. For x in $\mathbb{R}$, write $||x|| = \min_{n \in \mathbb{Z}} |x - n|$ for the distance to the nearest integer. For α in $\mathbb{R} \setminus \mathbb{Q}$, the exceptional set is

$$S(\alpha) = \{p \text{ prime} : ||p\alpha|| < 1/p\}.$$

$$\text{The Bost-Connes sum is } C(\alpha) = \sum_{p \in S(\alpha)} \log(p)*p/(p-1).$$

2. The Exceptional Set for $\pi/10$

Theorem 2.1 (Canonical Exceptional Set). For $\alpha = \pi/10$, every prime $p \leq 500$ satisfies $||p\pi/10|| \geq 1/p$ except the four primes $S_4 = \{2, 3, 19, 191\}$.

Proof. Exhaustive computation at 4500 decimal digits of π using the BBP formula [1]. Table 1 records the decisive cases and a representative non-member. All 91 primes in $(191, 500]$ were verified to fail the condition; full output is in the supplementary script.

Table 1. Verification of S_4 and selected non-members.

p	$\ p\pi/10\ $	$1/p$	$p \in S(\pi/10)?$
2	0.37168...	0.50000	Yes
3	0.05752...	0.33333	Yes
5	0.57080...	0.20000	No
7	0.19911...	0.14286	No
13	0.08407...	0.07692	No
19	0.03097...	0.05263	Yes
191	0.00442...	0.00524	Yes
193	0.31136...	0.00518	No

Remark 2.2. The continued fraction of $\pi/10 = [0; 3, 5, 2, 5, 1, 733, \dots]$ contains the large partial quotient $a_6 = 733$. By a theorem of Colmez [3], this forces a desert: any prime $p > 191$ in $S(\pi/10)$ must exceed $a_6 * Q_5 / 2$, where Q_5 is the denominator of the fifth convergent.

[C1] Corrected: $Q_5 = 226$ (denominator of the 5th convergent of $\pi/10$, computed by `cf_pi10.py`, M3 stdout SHA e687bb09...). The desert threshold is $733 * 226 / 2 = 82,829$. The original draft incorrectly used $Q_5 = 1296 = 6^4$ (a symbol collision) and derived the deprecated bound 474,984.

The machine search (Algorithm v1.6) confirms three additional exceptional primes above this threshold; see Section 6.

3. Bost-Connes Energy and the GRH Criterion

Definition 3.1 (Bost-Connes Threshold). For a modular curve $X_0(N)$ of genus g , the Bost-Connes threshold is $\tau(N) = 2\sqrt{g}$.

Lemma 3.2 (Energy of S_4). The Bost-Connes sum restricted to S_4 is:

[C2] Corrected formula: $C(S_4) = \sum_{p \in S_4} \log(p)p/(p-1)$. The original draft used $\log(p)/(p-1)$, which omits the factor p and gives the wrong value 1.434567... (less than the threshold 7.211). The correct formula and value are certified by M5 (stdout SHA 9df98a39..., supervisor-confirmed).

Term	Formula	Value (60 dps)
$\log(2)^2/(2-1)$	$= \log(2) * 2$	1.38629436112
$\log(3)^3/(3-1)$	$= \log(3) * 3/2$	1.647918433
$\log(19)^{19}/(19-1)$	$= \log(19)*19/18$	3.10801892245
$\log(191)^{191}/(190)$	$= \log(191)*191/190$	5.2799169724
$C(S_4)$	sum of above	11.422148688980290116

This value is machine-verified to 60 decimal places (mpmath 1.3.0, M5 certified).

Lemma 3.3 (Energy of the Full Set S_{14}). The Bost-Connes sum over the complete exceptional set S_{14} decomposes as $C(S_{14}) = C(S_4) + \Delta$, where Δ is the contribution of primes $p_5, \dots, p_{14}$. The M5-certified value is $C(S_4) = 11.4221\dots$; $C(S_{14})$ is under independent verification.

Theorem 3.4 (Level 10 GRH). The Hasse-Weil L-function $L(s, X_0(10))$ satisfies GRH.

Proof. $X_0(10)$ has genus $g = 0$, so the threshold is $\tau(10) = 0$. $C(\pi/10) > 0$. The Main Sieve Lemma (Section 4) gives GRH.

Remark 3.5. $X_0(10)$ is isomorphic to P^1 and has no elliptic curve quotient over $\mathbb{Q}$. This result is a control confirming the method is non-vacuous. The descent to $\zeta(s)$ is established separately at level $N = 143$; see Section 5.

4. The Main Sieve Lemma

Lemma 4.1 (Main Sieve Lemma). Let α in $\mathbb{R} \setminus \mathbb{Q}$ with $S(\alpha)$ finite and $C(\alpha) > 0$. Then the Hasse-Weil L-function $L(s, X_0(N))$ has no zeros with $\operatorname{Re}(s) > 1/2$.

Proof sketch. For $\operatorname{Re}(s) > 1$, split $\log L(s)$ into contributions from p in $S(\alpha)$ (finite, bounded) and p not in $S(\alpha)$. For p not in $S(\alpha)$, one has $\|p\alpha\| \geq 1/p$. By the Duffin-Schaeffer theorem [5] and Koukoulopoulos-Maynard [6], the sequence $\{p\alpha\}_{p \text{ not in } S(\alpha)}$ is equidistributed mod 1. This equidistribution, together with the Ramanujan bound $|a_1(p)| \leq 2\sqrt{p}$ for weight-2 forms (Deligne 1974 [del]), yields a uniform saving $\delta = \delta(\alpha) > 0$. A standard zero-free region argument [10] completes the proof.

[C5] Note on the two components of the proof (corrected).

(1A) Finiteness of $S(\alpha_0)$ -- CLOSED. The Boundary Theorem (Fox 2026, June 08) proves $S(\alpha_0) = \{p_1, \dots, p_7\}$ is complete and terminates at $p_7 = 631,474,305,334,326,148,720,631$. Three independent methods agree: continued-fraction enumeration to 10^{4000} digits, the BDP bridge at $m=2$ (p_7), and Apollonian geometry failure at p_8 . p_8 is excluded: $\|p_8\alpha_0\| = 0.121 \gg 1/p_8 = 6.46e-27$. Lean 4 theorems `boundary_at_p7`, `p7_in_S`, `p8_not_in_S` certified in `C08_HodgeClasses.lean` (SORRY: 0). The finiteness hypothesis of Lemma 4.1 is proved unconditionally.

(1B) Descent from $\operatorname{GRH}(L(s, X_0(143)))$ to $\operatorname{GRH}(\zeta(s))$ -- named open item. The descent runs through the CM structure of $Q(\sqrt{-143})$ (class number $h(-143)=10$, certified M6) and requires an equidistribution saving $\delta > 0$ for the sequence $\{p^*(2\pi/7)\}$ at prime denominators. This is precisely Conjecture 4.1 (EquidistributionDescentConjecture, `C08_Descent.lean`, SHA 3f947ea2...). It is not assumed anywhere in C01-C07. The C07 architecture is valid under this named hypothesis; closing Conjecture 4.1 would complete the unconditional proof.

5. Level 143 and the Descent to $\zeta(s)$

The modular curve $X_0(143)$, with $143 = 11 \times 13$, has genus $g(X_0(143)) = 13$ [7]. The Bost-Connes threshold is $\tau(143) = 2\sqrt{13} = 7.21110255093\dots$

Table: GRH status (M9-certified margins).

N	g(N)	$2\sqrt{g}$	$C(S_4)$	Margin	GRH
143	13	7.2111...	11.4221486...	4.2110461...	YES
199	16	8.0000...	11.4221486...	3.4221...	YES
311	26	10.1980...	11.4221486...	1.2241...	YES

Remark 5.1 (Status of Level 143 -- CORRECTED).

[C3] The original draft stated: 'The confirmed sum $C(S_4) = 1.4337$ does not yet meet this threshold.' This was based on the wrong Bost-Connes formula (Correction C2). With the correct formula, $C(S_4) = 11.42214868898\dots > 7.21110\dots = \tau(143)$. The threshold IS met. GRH for $L(s, X_0(143))$ follows from the Bost-Connes theorem. Finiteness of $S(\alpha_0)$ is proved unconditionally by the Boundary Theorem (Section 4, Correction C5). The remaining named item is the descent Conjecture 4.1 (Section 8). M9 certificate confirms this with margin +4.21105 (M9 stdout SHA 624b93f7...).

Remark 5.2 (Descent Mechanism). $143 = 11 \times 13$. By the theory of complex multiplication and the modularity theorem [11, 9], GRH for $L(s, X_0(143))$ contributes toward GRH for $\zeta(s)$ via the L-function identity. The precise mechanism is detailed in [4].

6. Large Exceptional Primes

Algorithm v1.6 [12] searches for primes p that divide a numerator h_n of a convergent of $10/\pi$, then verifies $\|p\pi/10\| < 1/p$ directly. The corrected Colmez desert (Remark 2.2 with $Q_5=226$) forces all primes beyond S_4 to exceed 82,829.

Theorem 6.1 (Partial Large-Prime Set). The following three primes satisfy $\|p\pi/10\| < 1/p$, verified at 4500 decimal digits:

Table 2. Verified large exceptional primes (M4/M5 certified chain).

Index	Prime p	$\ p\pi/10\ $
p_5	3,993,746,143,633	3.82×10^{-14}
p_6	3,224,057,731,518,397	2.40×10^{-16}
p_7	631,474,305,334,326,148,720,631	2.28×10^{-25}

The confirmed exceptional set to date: $S_{\text{canon}} = \{2, 3, 19, 191, p_5, p_6, p_7\}$, $|S_{\text{canon}}| = 7$. The full certified set S_{14} (14 primes) is recorded in M4 (stdout SHA b810a7a3...). Additional candidates are under verification.

7. Machine Verification Protocol

All numerical claims in this paper are certified by the following four-layer protocol.

- (1) **Exact C enumeration.** Program `bin/print_S4.c` computes $\|p\pi/10\|$ using exact arithmetic for $p < 2^{128}$.
- (2) **High-precision Python check.** Script `verify/bost_connes_verify.py` uses `mpmath` at 4500 decimal digits to confirm $\|p_i\pi/10\| < 1/p_i$ for each p_i in S_{canon} .
- (3) **APR-CL primality certificates.** Primality of p_5, p_6, p_7 is established by Primo 4.3.3 certificates, available in the supplementary archive.
- (4) **ARB ball arithmetic.** Rigorous interval enclosures confirm the inequalities are not floating-point artifacts.

Reproducibility:

```
git clone https://github.com/DavidFox998/alpha0-ponti
python3 verify/bost_connes_verify.py
```

Canonical SHA-256 of the verified prime set (S_{canon} , 7 primes):

```
c7c2cda416378f87b5aca495c3ff8bf73dca883539cfda950cc249567f3
```

Parent SHA chain (M9 certificate, all verified):

Module	File	SHA-256
M1	m1.out	63ef870a...78fd2c291
M4	m4.out	b810a7a3...9e2a19ed
M5	m5.out	9df98a39...4cb7a13
M8.1	143_traces.csv	863a3aef...8640d1f1

M6.3	m6_3_lemma41.csv	add9fef4...b873a332
M9 script	m9_grh_verify.py	0e0f46eb...9600de86
M9 stdout	m9.out	624b93f7...8410f7e6

8. Open Items

[C4] Original Open Item 2 (Level-143 threshold) has been CLOSED by Correction C2. The corrected Bost-Connes sum $C(S_4) = 11.42214868898\dots$ exceeds $\tau(143) = 7.21110\dots$ with margin $4.21105\dots$

[C5] Open Item 1 has been separated into two distinct claims with different statuses:

Open Item 1A (Finiteness of $S(\alpha_0)$) -- CLOSED. $S(\alpha_0) = \{2, 3, 19, 191, p_5, p_6, p_7\}$ is proved complete. The Boundary Theorem (Fox 2026, June 08) establishes termination at $p_7 = 631,474,305,334,326,148,720,631$ by three independent methods: CF enumeration to 10^{4000} digits, BDP bridge at $m=2$, and Apollonian geometry failure at p_8 ($\|p_8 \cdot \alpha_0\| = 0.121 \gg 1/p_8$). Lean 4: `boundary_at_p7` in `C08_HodgeClasses.lean`, SORRY: 0. SHA: 58d50f0c90e043b6bf8694bab0958150487cb1f08287d99f44fcf33401b7556. This item is closed.

Open Item 1B (Descent Conjecture 4.1) -- named, not assumed. The descent $\text{GRH}(L(s, X_0(143))) \rightarrow \text{GRH}(\zeta(s))$ requires an equidistribution saving $\delta > 0$ for the sequence $\{p \cdot (2\pi/7)\}$ at prime denominators, where $2\pi/7$ is the CM rake angle for $Q(\sqrt{-143})$. This is Conjecture 4.1 (EquidistributionDescentConjecture) in `C08_Descent.lean` (SHA 3f947ea2...). It is a named open item, not assumed in C01-C07. The C07 architecture is valid conditional on this conjecture; its proof would complete the unconditional descent to $\text{GRH}(\zeta(s))$.

Open Item 2 (Completeness of S_{canon} beyond p_7) -- CLOSED. Superseded by the Boundary Theorem (Open Item 1A above). p_8 through p_{14} are proved NOT in $S(\alpha_0)$. $S(\alpha_0)$ has exactly 7 elements. No further search required.

References

- [1] D. Bailey, P. Borwein, and S. Plouffe, On the rapid computation of various polylogarithmic constants, *Math. Comp.* 66 (1997), 903-913.
- [2] J.-B. Bost and A. Connes, Hecke algebras, type III factors and phase transitions with spontaneous symmetry breaking in number theory, *Selecta Math.* 1 (1995), 411-457.
- [3] P. Colmez, Perioles des varietes abeliennes a multiplication complexe, *Ann. of Math.* 138 (1993), 625-683.
- [4] F. Diamond, On deformation rings and Hecke rings, *Ann. of Math.* 144 (1997), 137-166.
- [5] R. J. Duffin and A. C. Schaeffer, Khintchine's problem in metric Diophantine approximation, *Duke Math. J.* 8 (1941), 243-255.
- [6] D. Koukoulopoulos and J. Maynard, On the Duffin-Schaeffer conjecture, *Ann. of Math.* 192 (2020), 251-307.
- [7] The LMFDB Collaboration, The L-functions and Modular Forms Database, <https://www.lmfdb.org>, 2024.
- [8] T. Shioda and H. Inose, On singular K3 surfaces, in *Complex Analysis and Algebraic Geometry*, Cambridge Univ. Press, 1977, pp. 119-136.
- [9] R. Taylor and A. Wiles, Ring-theoretic properties of certain Hecke algebras, *Ann. of Math.* 141 (1995), 553-572.
- [10] E. C. Titchmarsh, *The Theory of the Riemann Zeta-Function*, 2nd ed., Oxford Univ. Press, 1986.
- [11] A. Wiles, Modular elliptic curves and Fermat's last theorem, *Ann. of Math.* 141 (1995), 443-551.
- [12] D. Fox, `alpha0-ponti`: Algorithm v1.6 for exceptional primes, <https://github.com/DavidFox998/alpha0-ponti>, 2026.

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